Program-1

<u>Aim</u>: Write a program of compute of rectangle, triangle, square, and circle.

Area of rectangle

l=float(input("enter the value l")) b=float(input("enter the value b"))
area=(l+b)
print("area of rectangle",area)

Output

enter the value l4
enter the value b5 area of rectangle 9.0

Area of perimeter

l=float(input("enter the value l"))
b=float(input("enter the value b")) peri=2*(l+b)
print("perrimeter of rectangle",peri)

Output

enter the value l4 enter the value b5 perimeter of rectangle 18.0

Area of triangle

a=float(input("enter the value a"))
b=float(input("enter the value b")) c=float(input("enter the value c")) triangle=a+b+c print("area of triangle",triangle)

Output

enter the value a4
enter the value b5 enter the value c6 area of triangle 15.0

Area of square

side=4
square=side*side print("area of square",square)

Output

area of square 16

Area of circle

pi=3.14 r=float(input("enter the value r")) areaofcircle=pi*r*r print("area of circle",areaofcircle)

Output

enter the value r7
area of circle 153.86

Program-2

<u>Aim</u>: Write a program to swap two numbers

  a. **Using third variable**
  b. **Without using third variable**

x=int(input("enter the value of x")) y=int(input("enter the value of y"))
print("before swapping number %d %d\n" %(x,y))
x=x+y y=x-y x=x-y
print("after swapping number %d %d\n" %(x,y))

Output

enter the value of x23 enter the value of y25

before swapping number 23 25 after swapping number 25 23

Program-3

Aim: Write a program to determine whether a number is positive, negative, or zero.

```
num = float(input("Input a number: "))  if
num > 0:
        print("It is positive number")
elif num == 0:
        print("It is Zero")
else:
        print("It is a negative number")
```

Output

Input a number: 89
It is positive number

Input a number: -99
It is a negative number
Input a number: 0 It
is Zero

Program-4

Aim Write a program to determine whether a given numbers is odd or even.

```
num = int (input ("Enter any number to test whether it is odd or even:"))
if (num % 2) == 0:              print ("The number is even") else:
print ("The provided number is odd")
```

Output

```
Enter any number to test whether it is odd or even:887
The provided number is odd
```

Program-5

Aim Write a program to compute roots of a quadratic equation.

```
import cmath

a =float(input("Enter the number")) b
= float(input("Enter the number")) c
= float(input("Enter the number")) d
= (b**2) - (4*a*c) sol1 = (-
bcmath.sqrt(d))/(2*a)
sol2 = (-b+cmath.sqrt(d))/(2*a)

print('The solution are {0} and {1}'.format(sol1,sol2))
```

Output

```
Enter the number1
Enter the number5
Enter the number6
The solution are (-3+0j) and (-2+0j)
```

Program-6

Aim Write a program to find the number of and sum of all integers greater than 500 and less than 1000 that are divisible by 6.

```
Count = 0
Total_sum = 0
For num in range(501, 1000):
   If num % 6 == 0:
      Count += 1
      Total_sum += num
Print("Number of integers divisible by 6:", count)
Print("Sum of integers divisible by 6:", total_sum)
```

Output:

Number of integers divisible by 6: 83
Sum of integers divisible by 6: 62250

Program-7

Aim Write a program that takes two integer operands and one operator(+,- ,/,%,*) from the user, perform the requested operation and output the results.

```
operand1 = int(input("Enter the first integer operand: "))

operand2 = int(input("Enter the second integer operand: "))

operator = input("Enter the operator (+, -, *, /, %): ") result = 0

if operator == '+':

   result = operand1 + operand2  elif
operator == '-':
   result = operand1 - operand2 elif

operator == '*':

   result = operand1 * operand2

elif operator == '/':

if operand2 != 0:

      result = operand1 / operand2

else:          print("Error: Division

by zero") elif operator == '%':     if

operand2 != 0:

      result = operand1 % operand2

else:         print("Error: Modulo by zero")

else:

   print("Invalid operator")
print("Result: ", result)
```

Output:

Enter the first integer operand: 4

Enter the second integer operand: 3

Enter the operator (+, -, *, /, %): +   Result:

7

Program-8

Aim Given three numbers, write a program to determine whether they can from sides of an isosceles triangle.

A = float(input("Enter the first number: "))

B = float(input("Enter the second number: "))

C = float(input("Enter the third number: ")) If a == b and a != c:

Print("Yes, these numbers can form an isosceles triangle.")

Elif a == c and a != b:

Print("Yes, these numbers can form an isosceles triangle.") Elif b

== c and b != a:

Print("Yes, these numbers can form an isosceles triangle.")

Else: Print("No, these numbers cannot form an isosceles

triangle.") **Output:**

Enter the first number: 4
Enter the second number: 2
Enter the third number: 6 No, these numbers cannot
form an isosceles triangle.

Program-9

Aim Given three points A(x1,y1),B(x2,y2),C(x3,y3), write a program to determine whether they are collinear i.e., lie on the same line or not.

X1, y1 = map(float, input("Enter coordinates of point A (x1 y1):

").

split()) X2, y2 = map(float, input("Enter coordinates of point B

(x2 y2): ").split()) X3, y3 = map(float, input("Enter coordinates of

point C (x3 y3): "). split()) If (x1 * (y2 – y3) + x2 * (y3 – y1) +

x3 * (y1 – y2)) == 0:

Print("These points are collinear and lie on the same line.")

Else:

Print("These points are not collinear.")

Output:

Enter coordinates of point A (x1 y1): 2 3
Enter coordinates of point B (x2 y2): 4 3
Enter coordinates of point C (x3 y3): 5 3 These
points are collinear and lie on the same line.

Program-10

<u>Aim</u> Write a program to determine whether a number is palindrome or not.

```
i=int(input("Enter Number"))
rev=0 x=i while(i>0):
rev=(rev*10)+i%10
i=i//10 if(x==rev):
print("palindrome number") else:
   print("not palindrome")
```

Output

Enter Number454 palindrome
number

Program-11

<u>Aim</u> Write a program to determine whether a year is prime or not.

```
number = int(input("Enter any number: ")) if
number > 1:     for i in range(2, number):
if (number % i) == 0:             print(number,
"is not a prime number")            break     else:
print(number, "is a prime number") else:
print(number, "is not a prime number")
```

Output

Enter any number: 13

13 is a prime number.

Program-12

<u>Aim</u> Write a program to enter a number and calculate the sum of its digits

```
def getSum(n):
sum = 0      for
digit in str(n):
     sum += int(digit)
   return sum

n =int(input("Enter the number"))  print(getSum(n))
```

Output

Enter the number12345
15

Program-13

Aim Write a program to find the sum of first n positive integer numbers.

```
num= int(input("Enter the number"))
sum= 0 for i in
range(num+1):   sum+=i
print(sum)
```

Output

Enter the number8
36

Program-14

Aim Write a program to print Fibonacci series,i.e.,11235813.

```
nterms = int(input("How many terms? "))

# first two terms n1,
n2 = 0, 1
count = 0

# check if the number of terms is valid if
nterms <= 0:
   print("Please enter a positive integer")
# if there is only one term, return n1 elif
nterms == 1:
   print("Fibonacci sequence upto",nterms,":")
print(n1)
# generate fibonacci sequence else:
   print("Fibonacci sequence:")
while count < nterms:
       print(n1)
nth = n1 + n2        #
update values
n1 = n2
       n2 = nth        count +=
1Program-15
```

Output:

How many terms? 17 Fibonacci
sequence: 0
1
1
2
3
5
8

13
21
34
55
89
144
233
377
610
987

Program-15

<u>Aim</u> Write a program to compute factorial of number.

```
num = 7  factorial = 1 if num < 0: elif num == 0:
   print("The factorial of 0 is 1")
else:    for i in range(1,num + 1):
factorial = factorial*i     print("The factorial
of",num,"is",factorial)
```

Output

The factorial of 7 is 5040

Program-16

<u>Aim</u> :Write a program to compute determine whether is number is Armstrong?

```
num = int(input("Enter a number: "))
sum = 0 temp = num while temp > 0:
digit = temp % 10    sum += digit ** 3
temp //= 10 if num == sum:
print(num,"is an Armstrong number") else:
   print(num,"is not an Armstrong number")
```

Output

Enter a number: 663
663 is not an Armstrong number

Program-17

<u>Aim</u> :Write a program to compute HCF of a given numbers.

```
def compute_hcf(x, y):
   if x > y:
 smaller = y
else:
smaller = x      for i in range(1,
smaller+1):
 if((x % i == 0) and (y % i == 0)):
hcf = i
```

return hcf
num1 = 54   num2
= 24
print("The H.C.F. is", compute_hcf(num1, num2))

Output Program-18

The H.C.F. is 6

Aim: Write a program to read the age of 100 persons and count the number of persons in the age group of 50 to 60.

Initialize a variable to count the number of persons in the age group

[50,60]count=0

Input ages for 100 persons for
i in range(1, 10):

    age = int(input(f"Enter the age of person {i}: "))   # Check
if the age is in the range [50, 60)   if age >= 50 and age < 60:

        count += 1

Print the count of persons in the age group [50, 60) print(f"The number of persons in the age group [50, 60) is: {count}")

Output:

Enter the age of person 1:
Enter the age of person 2:
Enter the age of person 3:
Enter the age of person 4:
Enter the age of person 5:
Enter the age of person 6:
Enter the age of person 7:
Enter the age of person 8:
Enter the age of person 9:
The number of persons in the age group [50 , 60] is: 2

Program-19

Aim: Write a program to read a positive integer and determine and print its binary equivalent.

Input a positive integer from the user num =

int(input("Enter a positive integer: ")) # Check if the

entered number is positive if num <= 0:

print("Please enter a positive integer.") else:

    # Convert the integer to its binary equivalent and remove the '0b' prefix

binary_representation = bin(num)[2:]   # Print the binary representation

print(f"The binary representation of {num} is:

{binary_representation}")

Output:

Enter a positive integer: 78
The binary representation of 78 is: 100110

Program-20

Aim:Write a program to calculate pow(x,n),i.e., to calculate x”.

```
def power(x, n):      pow = 1      for i
in range(n):          pow
= pow * x

    return pow if __name__ ==
'__main__':

    x = 2     n = 3
print(power(x, n))
```

Output

8

Program-21

Aim: Count occurrence of a digit 5 in a given integer number input by the user.

```
def countOccurrences(n, d):
    # Initialize dictionary with keys 0 to 9 and values 0
digitCounts = {i: 0 for i in range(10)}

    # Convert number to string and loop through its digits
numString = str(n)
    for digit in numString:
digitCounts[int(digit)] += 1

    # Return count for the digit we are interested in
return digitCounts[d] if __name__ ==
"__main__":
    d = 2     n = 214215421
print(countOccurrences(n, d))
```

Output:

3

Program:22

Aim: Print geometric and harmonic means of a series input by the user.

```
from statistics import geometric_mean, harmonic_mean, mean

print(mean([1, 3, 7, 2, 5]))

print(geometric_mean([1, 3, 7, 2, 5]))

print(harmonic_mean([1, 3, 7, 2, 5]))
```

Output:

```
3.6
2.913693458576192
2.297592997811816
```

Program:23

Aim: Evaluate the following expressions.

a. **$x-x^2/2+x^3/4!+\ldots x^3!$**

b. **$x-x^3/3!+x^5/5!-x^7!+\ldots x^n/n!$**

```
def fact(m):   if m == 0:
return 1   else:
return m*fact(m-1) s=0
x=int(input("enter x:"))
n=int(input("enter n:"))
for i in range(1,n+1):

    s= s + ((-1)**(i+1))*((x**i)/fact(i))
print(s)
```

```
def fact(m):   if m == 0:
return 1   else:
return m*fact(m-1) s=0
x=int(input("enter x:"))
n=int(input("enter n:"))
for i in range(1,n+1):
    k= 2*i - 1
    s= s + ((-1)**(i+1))*((x**k)/fact(k))
    print(s)
```

Output:

enter x:1 enter
n:3
1.0
0.5
0.6666666666666666

= RESTART: C:/Users/ACER/AppData/Local/Programs/Python/Python311/142.py
enter x:1 enter n:4
1.0
0.8333333333333334
0.8416666666666667
0.841468253968254

Program:24

Aim: Print all possible combination of 4,5and6

```
import itertools as IT a=
int(input("enter 1st value")) b=
int(input("enter 2nd value")) c=
int(input("enter 3rd value"))
A= IT.combinations([a,b,c],2)
for i in A:     print (i)
```

Output:

enter 1st value4
enter 2nd value5
enter 3rd value6
(4, 5)
(4, 6)
(5, 6)

Program:25

Aim: Determine prime numbers within a specific range.

```
a = int(input("Enter START: "))
b = int(input("Enter END: "))
primes = []
for num in range(a,b+1):
 for p in range(2,num):
  if(num%p == 0):
    break
 else:
  primes.append(num)
for i in primes:
 print(i, end=',')
```

OUTPUT:

Enter START: 3

```
Enter END: 9
3,5,7,
```

Program:26

Aim: Compute transpose of a matrix.

```
X = [[12,7],
    [4 ,5],
    [3 ,8]]

result = [[0,0,0],
         [0,0,0]]

# iterate through rows
for i in range(len(X)):
   # iterate through columns
   for j in range(len(X[0])):
       result[j][i] = X[i][j]

for r in result:
   print(r)
```

OUTPUT:

```
[12, 4, 3]
[7, 5, 8]
```

Program:27

Aim: Perform following operations on two matrices.

a. **Addition**
b. **Subtraction**
c. **Multiplication**

```
m1 = [[2,1,3],[2,5,3],[7,8,6]]
m2 = [[3,0,0],
    [0,3,0],
    [0,0,3]]
print("Addition : ")
print('[',end='')
for i in range(3):
  for j in range(3):
   print(m1[i][j] + m2[i][j], end=',')
  print()
print(']')
print("Subtraction : ")
print('[',end='')
for i in range(3):
  for j in range(3):
```

```
    print(m1[i][j] - m2[i][j], end=',')
  print()
print(']')
print("Multiplication : ")
result = [[0,0,0],[0,0,0],[0,0,0]]
for i in range(3):
 for j in range(3):
  for k in range(3):
    result[i][j] += m1[i][k] * m2[k][j]
for i in result:
 print(i)
```

OUTPUT:

```
Addition :
[5,1,3,
2,8,3,
7,8,9,
]
Subtraction :
[-1,1,3,
2,2,3,
7,8,3,
]
Multiplication :
[6, 3, 9]
[6, 15, 9]
[21, 24, 18]
```

Program:28

Aim: Count total number of vowels in a word.

```
word = input(“Enter Any Word: ”)
vowels = 'aeiou'
count=0
for letter in word:
  if letter in vowels :
     count+=1
print(f'Vowels in {word} are {count}')
```

OUTPUT:

```
Enter Any Word: Hello World
Vowels in Hello World are 3
```

Program:29

Aim: Determine whether a string is palindrome or not.

```
word = input("Enter Any Word: ")
is_palin = True
for i in range(int(len(word)/2)):
```

```
  if(word[i] != word[-i-1]):
    is_palin = False
    break

if(is_palin):
  print(f'{word} is a Palindrome')
else:
  print(f'{word} is not a Palindrome')
```

OUTPUT:

```
Enter Any Word: MALAYALAM
MALAYALAM is a Palindrome
```

Program:30

Aim: Perform following operations on a list of numbers:

d. Insert an element

e. delete an element

f. sort the list

g. delete entire list

```
nums = [4,5,7,6,1,3,2]

insert = int(input("Enter Number to Insert: "))
nums.append(insert)
print(f'New List: {nums}')

delete = int(input("Enter Number to Delete: "))
nums.remove(delete)
print(f'New List: {nums}')
nums.sort()
print(f'Sorted List: {nums}')

print("Deleting entire list")
del(nums)
```

OUTPUT:

```
Enter Number to Insert: 8
New List: [4, 5, 7, 6, 1, 3, 2, 8]

Enter Number to Delete: 5
New List: [4, 7, 6, 1, 3, 2, 8]

Sorted List: [1, 2, 3, 4, 6, 7, 8]
Deleting entire list
```

Program:31

Aim: Display word after sorting in alphabetical order.

```
word = input("Enter a word: ")

# Sort the letters in the word in alphabetical
order
sorted_word = ''.join(sorted(word))

# Display the sorted word
print("Word after sorting in alphabetical order:",
sorted_word)
```

OUTPUT:

```
Enter a word: programming
Word after sorting in alphabetical order:
aggimmnoprr
```

Program:32

Aim: Perform sequential search on a list of given numbers.

```
def sequential_search(numbers, target):
    for i, num in enumerate(numbers):
        if num == target:
            return i  # Return the index where the
target is found
    return -1  # Return -1 if the target is not found

# Input a list of numbers
numbers = input("Enter a list of numbers
separated by spaces: ")
numbers = [int(x) for x in numbers.split()]

# Input the target number to search for
target = int(input("Enter the number to search
for: "))

# Perform the sequential search
index = sequential_search(numbers, target)

# Display the result
if index != -1:
    print(f"The target number {target} is found at
index {index}.")
else:
```

```
    print(f"The target number {target} is not in
the list.")
```

OUTPUT:

Enter a list of numbers separated by spaces: 5 7 8 4 3 2 1 6
Enter the number to search for: 8
The target number 8 is found at index 2.

Program:33

Aim: Perform sequential search on ordered list of given numbers.

```
def sequential_search(numbers, target):
    for i, num in enumerate(numbers):
        if num == target:
            return i  # Return the index where the
target is found
    return -1  # Return -1 if the target is not found

# Input a list of numbers
numbers = input("Enter a list of numbers
separated by spaces: ")
numbers = [int(x) for x in numbers.split()]

# Input the target number to search for
target = int(input("Enter the number to search
for: "))

# Perform the sequential search
index = sequential_search(numbers, target)

# Display the result
if index != -1:
    print(f"The target number {target} is found at
index {index}.")
else:
    print(f"The target number {target} is not in
the list.")
```

OUTPUT:

Enter an ordered list of numbers separated by spaces: 1 2 3 4 5 6
Enter the number to search for: 5
The target number 5 is found at index 4.

Program:34

Aim: Perform all operations of sets such as union, intersection, difference, symmetric difference.

```
# sets are define
A = {0, 2, 4, 6, 8};
B = {1, 2, 3, 4, 5};

# union
print("Union :", A | B)

# intersection
print("Intersection :", A & B)

# difference
print("Difference :", A - B)

# symmetric difference  print("Symmetric
difference :", A ^ B)
```

Output:

```
('Union :', set([0, 1, 2, 3, 4, 5, 6, 8]))
('Intersection :', set([2, 4]))
('Difference :', set([8, 0, 6]))
('Symmetric difference :', set([0, 1, 3, 5, 6, 8]))
```

Program:35

AIM: Maintain a record of books as per their serial numbers in library using python dictionary.

```
library = {}
# Function to add a book to the library def
add_book(serial_number, title, author):
 if serial_number in library:
print("Book with this serial number already exists.")
else:
       book_info = {"Title": title, "Author": author}
library[serial_number] = book_info
       print("Book added to the library.")

# Function to display book details by serial number
def display_book(serial_number):
    if serial_number in library:
book_info = library[serial_number]
print(f"Serial Number: {serial_number}")
print(f"Title: {book_info['Title']}")
print(f"Author: {book_info['Author']}")
```

```
else:
        print("Book not found in the library.")

# Function to remove a book from the library by serial number
def remove_book(serial_number):     if serial_number in
library:        del library[serial_number]        print("Book
removed from the library.")     else:
        print("Book not found in the library.")

# Main program loop while True:
print("\nLibrary Management System")
    print("1. Add a book")
print("2. Display book details")
print("3. Remove a book")
    print("4. Exit")     choice = input("Enter your choice:
") if choice == "1":    serial_number = int(input("Enter
the serial number: "))
  title = input("Enter the title: ")
author = input("Enter the author: ")
        add_book(serial_number, title, author)

    elif choice == "2":        serial_number = int(input("Enter the serial number of the book
you want to display: "))        display_book(serial_number)

    elif choice == "3":        serial_number = int(input("Enter the serial number of the book
you want to remove: "))        remove_book(serial_number)

    elif choice == "4":
        break
```

OUTPUT:

```
Enter your choice: 1
Enter the serial number: 254
Enter the title: Python
Enter the author: Dr.Pooja Sharma Book
added to the library.

Library Management System
1. Add a book
2. Display book details
3. Remove a book
4. Exit
Enter your choice: 2
Enter the serial number of the book you want to display: 254
Serial Number: 254
Title: Python
Author: Dr.Pooja Sharma
```

Library Management System
1. Add a book
2. Display book details
3. Remove a book
4. Exit
Enter your choice: 3
Enter the serial number of the book you want to remove: 254
Book removed from the library.

Library Management System
1. Add a book
2. Display book details
3. Remove a book
4. Exit
Enter your choice: 4

Program:36

Aim: Perform following operations on dictionary

1) Insert 2) delete 3) change

```
my_dictionary = {"name": "John","age": 30,"city": "New York"}

# Function to insert a key-value pair into the dictionary
def                 insert_key_value(key,                 value):
my_dictionary[key] = value
   print(f"Inserted: '{key}': '{value}'")

# Function to delete a key from the dictionary
def delete_key(key):     if key in
my_dictionary:         del my_dictionary[key]
print(f"Deleted: '{key}'")     else:
      print(f"Key '{key}' not found in the dictionary")

# Function to change the value of a key in the dictionary
def change_value(key, new_value):     if key in
my_dictionary:         my_dictionary[key] = new_value
print(f"Changed: '{key}' to '{new_value}'")
   else:
      print(f"Key '{key}' not found in the dictionary")

# Main program loop while True:
print("\nDictionary Operations")
   print("1. Insert")     print("2.
Delete")     print("3. Change")
print("4. Display Dictionary")
print("5. Exit")

   choice = input("Enter your choice: ")
```

```
    if choice == "1":         key =
input("Enter the key to insert: ")
   value = input("Enter the value: ")
        insert_key_value(key, value)

    elif choice == "2":         key =
input("Enter the key to delete: ")
delete_key(key)

    elif choice == "3":         key = input("Enter
the key to change: ")         new_value =
input("Enter the new value: ")
        change_value(key, new_value)

    elif choice == "4":         print("Current
Dictionary:", my_dictionary)

    elif choice == "5":
        break
```

OUTPUT:

Dictionary Operations
1. Insert
2. Delete
3. Change
4. Display Dictionary 5. Exit
Enter your choice: 1
Enter the key to insert: Name
Enter the value: Drishti
Inserted: 'Name': 'Drishti'

Dictionary Operations
1. Insert
2. Delete
3. Change
4. Display Dictionary
5. Exit
Enter your choice: 2
Enter the key to delete: city
Deleted: 'city'

Dictionary Operations
1. Insert
2. Delete
3. Change
4. Display Dictionary
5. Exit
Enter your choice: 3

Enter the key to change: Name
Enter the new value: Shrishti
Changed: 'Name' to 'Shrishti'

Dictionary Operations
1. Insert
2. Delete
3. Change
4. Display Dictionary
5. Exit
Enter your choice: 4
Current Dictionary: {'name': 'John', 'age': 30, 'Name':'Shrishti'}

Dictionary Operations
1. Insert
2. Delete
3. Change
4. Display Dictionary 5. Exit
Enter your choice: 5

Program:37

Aim: Check whether a number is in a given range using functions.

```
def test_range(n):     if n in range(3,9):
print( " %s is in the range"%str(n))
else :
      print("The number is outside the given range.")
test_range(5)
```

Output:

5 is in the range

Program:38

Aim: Compute area of different shapes such as circle, rectangle, square, triangle,rhombus, parallelogram, cone , cylinder , cube cuboid, sphere , etc. using user defined functions.

```
#Area of different shapes def
area(shape):
  shape=shape.lower()   if(shape=="circle"):
rad=float(input("Enter radius:"))      return 3.14*rad**2
elif(shape=="square"):      side=float(input("Enter Side:
"))      return side**2   elif(shape=="triangle"):
base=float(input("Enter Base:"))
height=float(input("Enter Height: "))     return
0.5*base*height    elif(shape=="rectangle"):
length=float(input("Enter Length: "))
width=float(input("Enter Width: "))     return
2*(length+width)   elif(shape=="rhombus"):
```

```
d1=float(input("Enter First Diagonal:"))
d2=float(input("Enter Second Diagonal:"))
return 0.5*d1*d2    elif(shape=="cone"):
r=float(input("Enter Base of cone:"))
sl=float(input("Enter Slant Height:"))     return
3.14*r*(r+sl) print("Compute Area of Different
Shapes\n1.Circle\n2.
Square\n3. triangle\n4. Rectangle\n5.Rhombus\n6
    Cone")
shp=input("Enter Choice(name): ")
print("Area       is:       ",       area(shp))
```

OUTPUT:

```
Compute Area of Different Shapes
1. Circle
2. Square
3. triangle
4. Rectangle
5. Rhombus
6. Cone
Enter Choice(name): square
Enter Side: 5
Area is:  25.0
```

Program:39

Aim: Design a scientific calculator which performs all arithmetic functions, trigonometric functions, random numbers functions, and mathematical functions.

```
import math  import
random
print("-----CALCULATOR------")
print("1. Arithmetic")
print("2. Trignometric")
print("3. Random Number")
print("4. Mathematic")

def arithmetic(string):   num1 =
int(input("Enter First Number: "))    num2 =
int(input("Enter Second Number: "))
if(string == 'plus'):     return num1+num2
elif(string == 'minus'):     return num1-num2
elif(string == 'mul'):     return num1*num2
elif(string == 'div'):     return num1/num2 def
trignometric(string):      angle =
int(input("Enter Degrees: "))      if(string ==
'sin'):
      return math.sin(math.radians(angle))
elif(string == 'cos'):
```

```
        return math.cos(math.radians(angle))
    elif(string == 'tan'):
        return math.tan(math.radians(angle)) choice
    = int(input("Enter Your choice(1/2/3/4): "))
    if(choice==1):
      print("Options: \n1. Addition\n2. Subtraction\n3.    Multiplication\n4.
Division")    opt = int(input("Enter Your
    Option(1/2/3/4): "))    if(opt==1):
        print('Reuslt is: ', arithmetic('plus'))
    elif(opt==2):
        print("Result is: ", arithmetic('minus'))
    elif(opt==3):
        print("Result is: ", arithmetic('mul'))
    elif(opt==4):
        print("Result is: ", arithmetic('div'))
    if(choice==2):   print("Options: \n1. Sine\n2.
    Cosine\n3. Tangent")   opt = int(input("Enter Your
    Option(1/2/3): "))    if(opt==1):
      print('Reuslt is: ', trignometric('sin'))
    elif(opt==2):
      print("Result is: ", trignometric('cos'))
    elif(opt==3):
      print("Result is: ", trignometric('tan'))
    if(choice==3):
     print("Generating Random Number:",
    random.randint(0,99))  if(choice==4):     print("Options: \n1.
    Square Root\n2. Exponent")     opt = int(input("Enter Your
    Option(1/2): "))     if(opt==1):       n1=float(input("Enter any
    number: "))
      print('Reuslt is: ', n1*0.5)
    elif(opt==2):       n1=float(input("Enter
    any number: "))        a=int(input("Enter
    Power: "))
      print("Result is: ", n1**a)
```

OUTPUT:

```
-----CALCULATOR------
1. Arithmetic
2. Trignometric
3. Random Number
4. Mathematic
Enter Your choice(1/2/3/4): 4 Options:
1. Square Root
2. Exponent
Enter Your Option(1/2): 1
Enter any number: 5
Reuslt is:  2.5
```

Program:40

Aim: Write a python function that accepts a string and calculates number of upper case letters and lower case letters available in that string.

```
def myfunc(string):
  uc = 0
  case_lc =
'abcdefghijklmnopqrstuvwxyz'    case_uc
= case_lc.upper()    lc = 0    for i in string:
if(i in case_lc):
     lc+=1    elif(i
in case_uc):
uc+=1    return
uc,lc
text = input("Enter any string: ")
print(f"Number of Upper cases and lower cases in {text} are {myfunc(text)}")
```

OUTPUT:

Enter any string: Python Programming

Number of Upper cases and lower cases in Python   Programming   are (2, 15)

Program:41

Aim: Find the max of three numbers using function.

```
def find_max(a,b,c):
if a>=b and b>=c:
return a    elif b>=a
and b>=c:       return
b    else:
     return c
a = int(input("Enter the Number")) b
= int(input("Enter the Number")) c
= int(input("Enter the Number"))
print("Max of Three Numbers is :",find_max(a,b,c))
```

Output:

Enter the Number56
Enter the Number58
Enter the Number59
Max of Three Numbers is : 59

Program-42

Aim: Multiply all the numbers in a list using functions.

```
#Multiply all the numbers in a list using functions.
def product(numbers):     total =1     for number in
numbers:
        total *=number
    return total

length = int(input("Enter list length"))
numbers_list =[] for _ in range(length):     temp =
int(input("Enter the numbers present in list :"))
    numbers_list.append(temp)


print(f"product of numbers present in list is :{product(numbers_list)}")
```

Output:

Enter list length:3
Enter the numbers present in list :1
Enter the numbers present in list :2
Enter the numbers present in list :5
product of numbers present in list is :10

Program-43

Aim: Compute mean, variance, and standard deviation of a given list of 20 numbers.

```
data = [2,3,4,6,7,9,11,16,19,22,24,26,30,31,34,35,38
                   ,40,42,45]
sum = 0  for
num in data:
sum+=num
mean = sum/20
print("Mean of Data is:", mean)
n_data = []  sum=0  i=0  for
num in data:
n_data.append((num-mean)**2)
 sum += n_data[i]  i+=1 var = sum/20
print("Population Variance of Data is:", var)
print("Standard Deviation of Data is:",
var**0.5)
```

OUTPUT:

```
Mean of Data is: 22.2
Population Variance of Data is: 194.36
Standard Deviation of Data is: 13.941
```

Program-44

Aim: Solve fibonacci sequence using recursion.

```
 def recur_fibo(n):
   if n <= 1:
return n    else:
       return(recur_fibo(n-1) + recur_fibo(n-2))

nterms = 8

# check if the number of terms is valid
if nterms <= 0:    print("Plese enter a
positive integer") else:
   print("Fibonacci sequence:")
for i in range(nterms):
print(recur_fibo(i))
```

Output:

```
Fibonacci sequence:
0
1
1
2
3
5
8
13
```

Program-45

Aim: Get the factorial of a non-negative integer using recursion.

```
def recur_factorial(n):    if n == 1:
  return n
else:
       return n*recur_factorial(n-1)

num = 9

# check if the number is negative if
num < 0:
   print("Sorry, factorial does not exist for negative numbers")
elif num == 0:
   print("The factorial of 0 is 1") else:
```

```
  print("The factorial of", num, "is", recur_factorial(num))
```

Output:

The factorial of 9 is 362880

Program-46

Aim: Write a program to create a module of factorial in python.

```
num = int(input("Enter a number: "))
factorial = 1     if
num < 0:
   print(" Factorial does not exist for negative numbers")
elif num == 0:        p
rint("The factorial of 0 is 1")
    else:
     for i in range(1,num + 1):
   factorial = factorial*i
  print("The factorial of",num,"is",factorial)
```

Output:

Enter a number: 10
The factorial of 10 is 3628800

Program-47

Aim: Design a python class named Rectangle,constructed by a length & width, also design a method which will compute the area of a rectangle.

```
class shape_rectangle(): def
__init__(self,my_length, my_breadth):
  self.length = my_length
self.breadth = my_breadth def
calculate_area(self):
  return self.length*self.breadth
len_val = 6 bread_val = 45
print("The length of the rectangle is : ")
print(len_val) print("The breadth of the
rectangle is : ") print(bread_val)
my_instance = shape_rectangle(len_val,bread_val)
print("The area of the rectangle : ")
print(my_instance.calculate_area())
print()
```

Output:

The length of the rectangle is :

6

The breadth of the rectangle is :

45

The area of the rectangle :

270

Program-48

Aim: Design a python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a circle.

```
class Circle:  def
__init__(self, radius):
  self.radius = radius
def compute_area(self):
  area = 3.14 * self.radius ** 2
return area  def
compute_perimeter(self):
  perimeter = 2 * 3.14 * self.radius
return perimeter
radius = float(input("Enter the radius of the circle:
")) circle = Circle(radius) area =
circle.compute_area() perimeter =
circle.compute_perimeter() print(f"The area of the
circle is: {area}") print(f"The perimeter of the circle
is: {perimeter}")
```

OUTPUT:

```
Enter the radius of the circle: 8
The area of the circle is: 200.96
The perimeter of the circle is: 50.24
```

Program-49

Aim: Design a python class to reverse a string 'word by word'.

```
class Reverse:
 def __init__(self, text):
  self.text = text  def reverse_words(self):
words = self.text.split()   reversed_words
= words[::-1]   reversed_text = "
".join(reversed_words)   return
reversed_text input_text = input("Enter a
string: ") reverser = Reverse(input_text)
reversed_text = reverser.reverse_words()
print("Reversed text:", reversed_text)
```

OUTPUT:

```
Enter a string: Python Programming
Reversed text: Programming Python
```

Program-50

Write a python program to read an entire text file.

```
file_path= "example.txt" with open(file_path, "r") as
 file:      file_content = file.read()
print("File Content:")
print(file_content)
```

OUTPUT:

```
File Content:
Hello World!!!
```

Program-51

Aim: Design a python program to read first n lines of a text file.

```
file_path = "example.txt" with open(file_path, "r") as file:
    n=int(input("Enter the number of lines to read: "))
   for i in range(n):
      line =
   file.readline()    if not
   line:      break
   print(line.strip())
```

OUTPUT:

```
Enter the number of lines to read: 4 1.
Hello World!!!
2. 3.
4.
```

Program-52

Aim: Construct a python program to write and append text to a file and display the text.

```
def file_read(fname):
 from itertools import islic with open(fname, "w") as myfile:
myfile.write("Python Exercises\n")
 myfile.write("Java Exercises")
 txt = open(fname)
   print(txt.read())
file_read('abc.txt')
```

Output:

Python Exercises Java
Exercises